

U-ESFM: What the current conclusions lack for an ICLR submission

Calibrated against the bar RESfM itself cleared at ICLR 2025

August 28, 2026

Calibration

RESfM was accepted to ICLR with single-seed results, no error bars, no adaptive mechanism, and OOD numbers on its own unreplicated track preprocessing. Judged by that revealed bar, most items below are *armor*, not entry requirements. The one structural difference: RESfM’s headline margins were large ($\sim 2\times$ over ESfM), so single-seed reporting was tolerable; **our headline is parity with small margins** (MegaDepth 0.316 vs. 0.344, an 8% gap, with measured seed std ± 0.08 – 0.09)—the claimed margins sit *inside* single-seed noise, which reviewers will catch. Hence one must-have, the rest optional strengtheners.

Must-have

1. **Seed error bars on the primary (finelr twin) table.** U-ESFM-finelr s21/s22: trained; the missing OOD/MD eval fleets (379 jobs, incl. Olsson) submitted Aug 28. RESfM-finelr s21/s22: *launched Aug 28* (jobs 400449/400451, ~ 2 days), our env (matches the table; env shown negligible for no-TTT evals). Goal: every main-table cell as mean $\pm$ std over 3 seeds; “on par without labels” becomes defensible.

Strongly advisable (each upgrades a caveat into a claim)

2. **TTT stack-sensitivity characterization.** New finding (Aug 28): under the authors’ pinned BLAS stack, no-TTT RESfM moves $0.344 \rightarrow 0.350$ (7/36 scenes) while U-ESFM weight+TTT moves $0.316 \rightarrow 0.453$ (20/36 scenes, bidirectional): TTT amplifies solver-level nondeterminism $\sim 20\times$. Same-env repeat control in flight to separate stack effect from TTT stochasticity. Needed: per-mechanism quantification, so the in-distribution parity claim is stated with its stability envelope rather than as a caveat.
3. **Author-track replication (external validity).** All OOD numbers are on our own rebuilds, explicitly not comparable to published numbers. Fadi Khatib’s preprocessed OOD tracks ($\sim$ Sep 4) unlock replication of the key arms on their preprocessing. (RESfM’s own OOD numbers are equally unreplicated—this is armor, but high-value.)
4. **Multi-dataset training under the canonical recipe.** The diversity finding (supervised gains 1DSfM but collapses clean OOD from epoch 1000; MADWEIGHT uniquely survives, BMVS 0.22 vs. 2.8–4.0) is currently confounded (5-milestone recipe, early-vs-late checkpoint regimes). Fix in flight: 9-run authors-config multids batch launched Aug 28 (U-ESFM at 10^{-4} , RESfM at both lrs, seeds 20–22, our env; jobs 410185–410213), then standard eval fleets.
5. **Learning-rate symmetry closure.** Base lr is per-method (10^{-4} self-sup, 10^{-3} supervised); disclosed, but the 2×2 is incomplete. In flight: RESfM at 10^{-4} under authors’ full recipe (faithful_lr1e4, job 389928).

Optional spine upgrade (would change the paper’s shape)

6. **Selector impossibility as a thesis.** The 4-way negative (contamination, reprojection, held-out CV, cycle consistency all fail; oracle headroom real at 87% decisive scenes) is currently “future work.” For ICLR it could be elevated to a systematic label-free-signal-wall analysis (demonstrated 3×: mechanism selection, checkpoint selection by OOD, run ranking)—the intellectual spine RESfM never attempted.
7. **Statistical treatment.** Per-scene bootstrap/significance on top of mean/median; expected by ICLR reviewers for small-margin claims.

Already exceeding the RESfM-revealed bar (no action)

Measured per-dataset contamination; the label-ceiling measurement (recall $0.82 \rightarrow 0.26\text{--}0.39$, coverage $99 \rightarrow 55\%$ above $\sim 15\%$); controlled twin protocol with full provenance ladder (scratch / faithful / matched-env / authors-code / released); regime-level adaptivity recommendation; environment noise-floor analysis.

Baseline coverage vs. the RESfM paper’s comparison set

RESfM (ICLR 2025) compares against ESfM, GASfM, Theia, GLOMAP, ESfM* (clean-track upper bound, unranked), COLMAP, VGGsFm, and MAST3R (the last two image-based, outside the shared-tracks protocol). Our coverage and the actions to close it:

- **ESfM** — done (author code on our tracks, shallow + deep, 5-seed OOD evals).
- **COLMAP** — done on our tracks (1DSfM-hard + spot checks); matched-env repeats running.
- **ESfM*** — *in flight (launched Aug 28)*: GT-4px-filtered “_star” track copies built for MegaDepth/1DSfM/1DSfM-hard/Strecha/BlendedMVS; ESfM trained and tested on clean tracks per the paper’s protocol. Highest-value missing baseline: it doubles as the oracle-removal upper bound for the label-ceiling argument and quantifies how much of that headroom label-free MADWEIGHT captures.
- **GLOMAP** — not computed. Next most valuable: modern classical global SfM, single binary, runs on COLMAP-format inputs; needed to preempt “RESfM compared to classical SfM, why don’t you?”
- **Theia** — not computed; same role as GLOMAP, older pipeline, lower priority.
- **GASfM** — blocked: forward/pre-BA works in our env, but post-BA requires their custom-compiled Ceres cost functors; closing it means wiring GASfM output into our pycolmap robust BA (engineering task).
- **VGGsFm / MAST3R** — lowest priority: image-based (outside the controlled shared-tracks protocol) and RESfM already reports they cannot run the large scenes.

Timeline anchors

RESfM-fineir seeds + faithful pair + lr probe + multids quartet: results within ~ 2 days of Aug 28. Author tracks: $\sim$ Sep 4. 3DV supplementary deadline: Sep 2 (no post-deadline results allowed). ICLR: all of the above land comfortably.